**Touchless gesture-controlled presentations**

for teachers, students, researchers and speakers

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## Beyond Touch

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# Shared clickers: lost, dead, dirty







## 81%

### Fecal bacteria found

on 81% of hotel-room surface samples, with TV remotes among the worst. Univ. of Houston, ASM meeting, 2012

## ~8×

### TV remote vs toilet seat

more germs on a remote than on a toilet seat. Compare the Market swab test

**Target Users:** teachers, students, surgeons and speakers

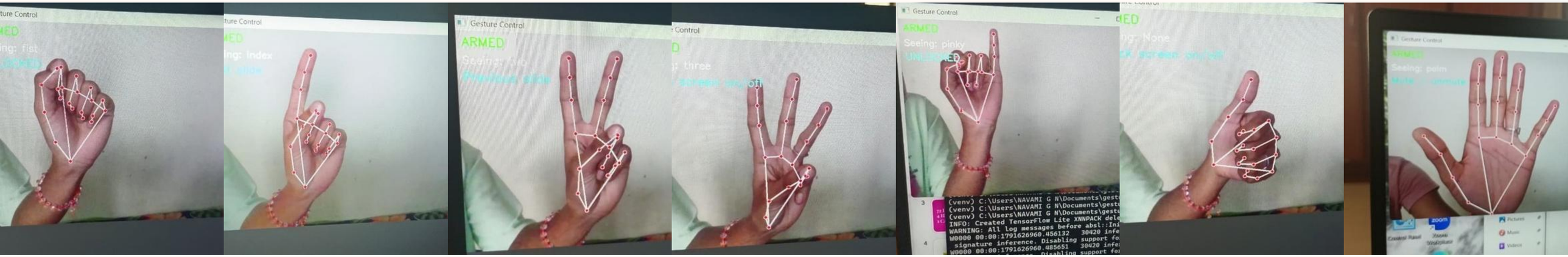
**Today:** clickers get lost, die, get shared and a mouse ties you to the desk

**Why gestures:** any webcam,  no contact, no battery

Figures are for shared handheld remotes in general, not clicker-specific studies.

# Your hand is the remote

Controls Google Slides and PowerPoint through the webcam




**01**

### Fist

Lock / unlock

**Hold 1.5 s**

**02**

### Index

Next slide

**Hold 0.4 s**

**03**

### Two fingers

Previous slide

**Hold 0.4 s**

**04**

### Three fingers

Black screen

**Hold 0.6 s**

**05**

### Pinky

End show

**Hold 0.8 s**

**06**

### Thumbs up

First slide

**Hold 0.6 s**

**07**

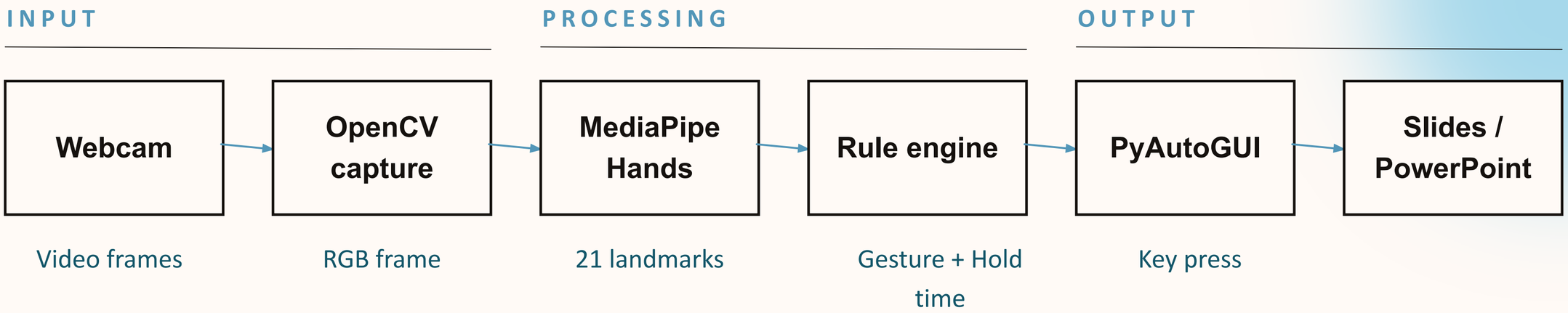
### Open palm

Mute / unmute

**Hold 1.0 s**

- **Nothing to lose**
- **Nothing to charge**
- **Nothing to touch**

# System design and tech stack




## Workflow

**01** Read fingers

**02** Recognise gesture

**03** Initiate Task

**04** 1 s cooldown

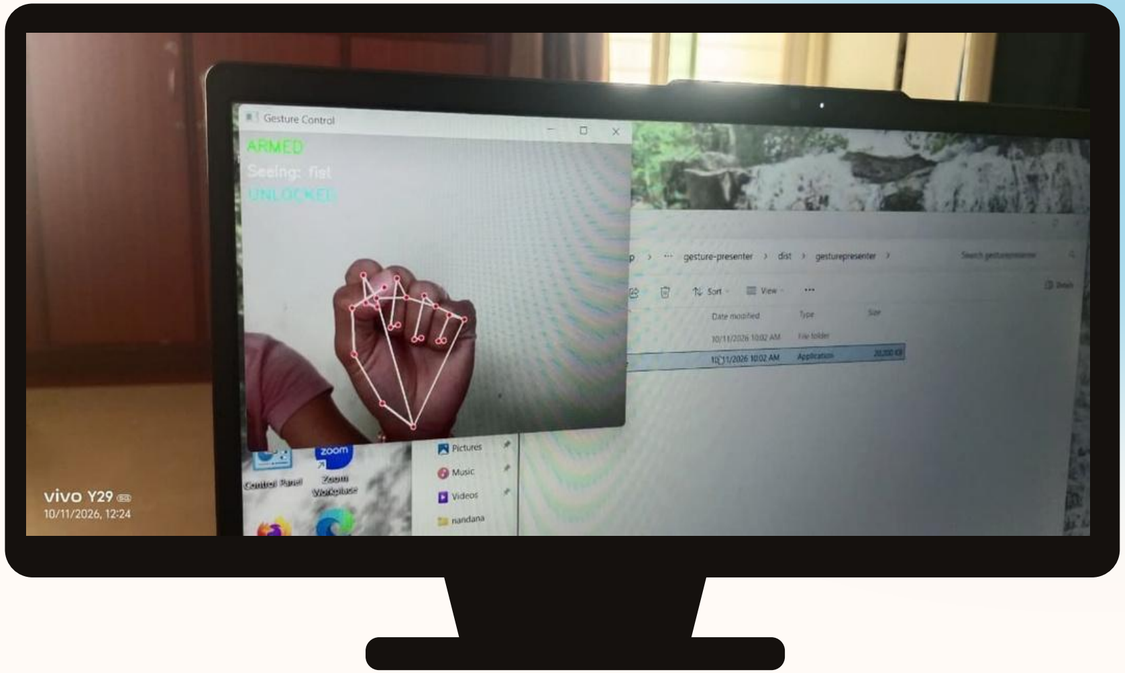
**21 landmarks per hand**

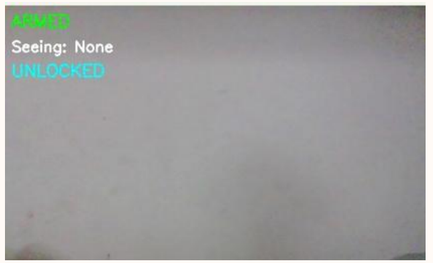
**95.7% palm-detection API**

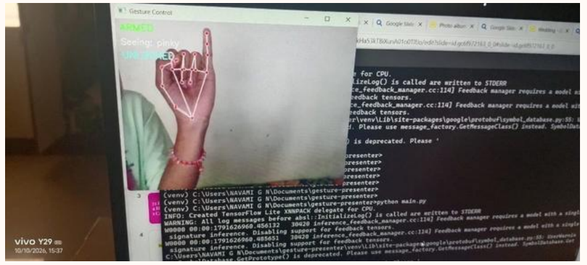
# Working prototype

## Measured results

| | |
|---|---|
| 90 % | Accuracy |
| 60 ms | Response time |
| 30+ FPS | Frame rate |
| 5 | Tests run |



Laptop + built-in webcam, no extra hardware

# It never misfires.

**Strongest differentiator:** A lock gesture, hold timers and a cooldown stop accidental movements from changing slides.

Building a hand-gesture presentation controller is a strong computer vision and human-computer interaction (HCI) project. By replacing physical clickers, laser pointers, and keyboards with vision-based hand tracking, you eliminate hardware dependencies and create a touchless experience.

## Unique features

- Fist lock / unlock
- Hold time per gesture
- Unclear gestures ignored

## Practicality

- ₹0 extra hardware
- Python, OpenCV, MediaPipe
- Live ARMED / LOCKED overlay

# Impact and future scope

**0**

Shared contact points

**₹0**

Extra hardware per user

**7**

Gestures which replace the clicker

**Zero Additional Hardware Cost (B2B SaaS Model)**

**Enhanced Presenter Mobility & Dynamics**

**Touchless Hygiene & Sterile Environments**

# THANK YOU

**Our Project is live on:** https://yashashmallesh.github.io/Beyond-touch/